

COURTSIDE: Calling the Point Before It Lands

Predictive Ball Tracking and the Value of Speed in In-Play Tennis Markets

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Abstract. We built a ball-tracking engine that calls a point (ball out or into the net) before it lands: on held-out table tennis it called 4 of 41 misses a median 162 ms early, none wrong, at 119.9 fps, and on broadcast tennis its out calls were right 5 of 5 at the bounce and 3 of 3 when 33 ms ahead. On 13,084 Polymarket ATP/WTa matches, wallets that trade within 3 s of a point earn after fees in all 11 months, as our pre-registered hypothesis predicted out of sample. What we could trade loses after costs: a copier of their trades, entering after the detection second, the venue's hold and the block-time lag, makes -1.42¢ a share in sample and -1.02¢ out of sample ($-\$31$, $-\$28$ a day; the OOS interval spans zero); a camera policy trading every point from our classifier's measured call statistics at an assumed licensed 0.5 s feed makes $-\$92$, $-\$146$ a day (upper-bound fills), less at 1 and 3 s, and earns only at a post hoc score-stamp delay ($+\$174$, $+\$65$). Untradable benchmarks price the speed we lack: the fast tier's own fills (v2; Sharpe 14.5, 6.7; doubled fees erase the OOS gain) and a computer-vision (CV) trader on points the market later repriced ($+\$15$ a day out of sample pre-registered, $+\$81$ post hoc). Our one test with rules fixed in advance holds the last 20% of matches, 10.7% of the calendar, so no untouched test meets the organizers' by-time rule.

JEL: G14, G12, C58. **Keywords:** computer vision; ball tracking; prediction markets; latency arbitrage; in-play betting; tennis.

1 Introduction and summary

A camera films the court; a model finds the ball, fits its flight and calls the point before the bounce (Fig. 1). A Polymarket price is the chance that a player wins and lags each point by about a second, so whoever learns the point first sets the price. We map who is fast, price each second of feed delay ($\$42\text{--}74$ a day for the first second, on points that later moved) and test what we could trade.

Why the benchmark Sharpe ratios exceed 3. v2's 14.5 comes from about 270 small bets a day placed just after points, held to resolution, with a mean same-day pairwise correlation of 0.0013, so daily P&L is rarely negative. It is priced at the fast tier's own fills, and 97.7% of its in-sample profit (99.2% OOS) is in prints stamped in the second the score move was detected: what their speed earns, not what a follower can (on later prints only: $+0.33\text{¢}$, $+0.76\text{¢}$ a share, both CIs spanning zero). Checking it, we found and fixed a detector start time that used later prices (in-sample Sharpe 16.8 to 14.5).

2 Economic hypothesis

A Markov model of the score gives the exact chance that each player wins from any score (Klaassen and Magnus, 2001), so each point moves fair value by a known amount at a known moment (the *swing*: mean 5.7%, 38.4% on a late break point). News reaches traders in tiers: court, video, then score feeds 28.2–44.1 s after the price moved. A maker whose quote rests after the point is picked off by whoever knows first (Glosten and Milgrom, 1985; Budish et al., 2015; Aquilina et al., 2022), as courtsiders do in

Rerun: bash reproduce.sh (<https://github.com/OjasMishra32/computervisionpredictivetabletennismoneyouuushiii>). Prices, fills, fees and venue delays are real Polymarket records; the CV trader's calls, their timing at an assumed feed latency and its fills on the measured book are simulated. Paper trading only: no real money, no licensed feed, no court camera.

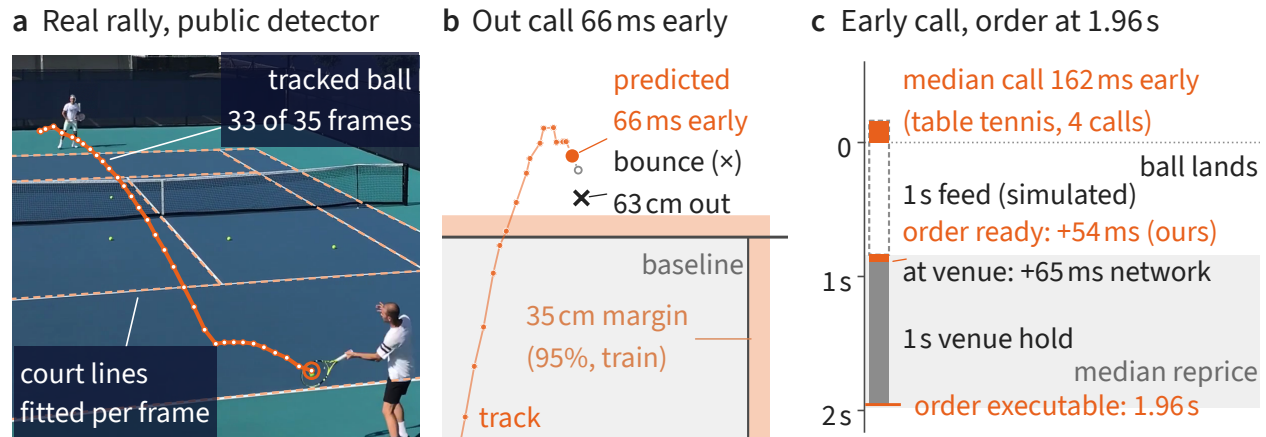


Figure 1. Three pieces of vision work, not one run. (a) Pexels stock rally: public TrackNet detector, our linking and court-line fit; ball in 33 of 35 frames. (b) Broadcast test bounce (detector trained on all ten games): out call 66 ms early at an assumed 25 fps; landing 69 cm off, ball (x) 63 cm out. (c) Eq. 1: the order comes 1,116 ms after the median reprice at the pre-registered lag, 24 ms before it post hoc.

betting markets (Croxson and Reade, 2014; Brown, 2012): on 482 live points a median \$222 of quotes rested at stale prices 0.25 s before the reprice (\$0 after 0.5 s), and Polymarket holds each taker order 1 s.

H6 (pre-registered). We expect wallets that trade within 3 s of a score move in Polymarket ATP and WTA match markets to earn a positive net *markout* over 30 s (profit at the price 30 s later, after fees), because the other side is a maker who has not yet cancelled a stale quote; the edge persists because physics and data licences decide who learns first. Wallets picked on past months then stay positive every month and out of sample; H6 fails if the markout is ≤ 0 in over a third of months, or out of sample. H1–H4 were committed before any result (7232986); H6, written after the in-sample tier results, was frozen before any held-out read.

Will the edge last? The fast tier’s net edge fell 0.16¢ a month (CI 0.06–0.25), mostly as the fee rose; a fee above 8.4% (today 5%) would end it.

3 Data and universe

Every resolved Polymarket ATP/WTA singles match market with at least \$5,000 traded: 13,084 matches (1,310 at Grand Slams; \$2.84B), Oct 2025–Oct 2026, from public APIs (Polymarket, 2026a), point in time, so no match is dropped by outcome: 378 settled 50/50 (302 never started) and 283 ended in a retirement, all at settlement price. *In sample* (IS) is the first 80% of matches by start time, *out of sample* (OOS) the last 2,617, from 2026-08-25 14:15 UTC, a match-count split pre-registered before any result (Section 8). Other data: Table A3 (Huang et al., 2019; Voeikov et al., 2020; Sackmann, 2026).

Data problems. Six tapes hit the API’s row cap (kept); 2,889 early matches had no published fee schedule (charged none); tape times are block times 1.98 s late; gaps are carried forward, never filled from later prints; no corporate actions apply. Regimes: the taker hold fell from 3 s to 1 s on May 15 and the fee rose from 3% to 5%; v2 is positive in each (+1.96¢ to +1.02¢ IS), and the CV trader is simulated only on the 7,755 matches with the 1 s hold.

4 Methodology

The fast tier and v2. Each month a wallet joins the fast tier if, in earlier months, it traded at least 30 times 0–3 s after a score move (a ≥ 4 ¢ move of the 10 s price from its prior 60 s) in 10+ matches (markout $t > 3$). v2 books their trades at their own fill prices, sized by resolution risk (best of 55 rules, walk-forward), with \$1,000 orders and 100 shares net per match, held to resolution. Our *copier* takes only v2 trades printed after the *detection second* (tapes stamp whole seconds, unordered within one) and fills after that print, the network leg, the venue’s hold and the median 1.98 s block-time lag, at the

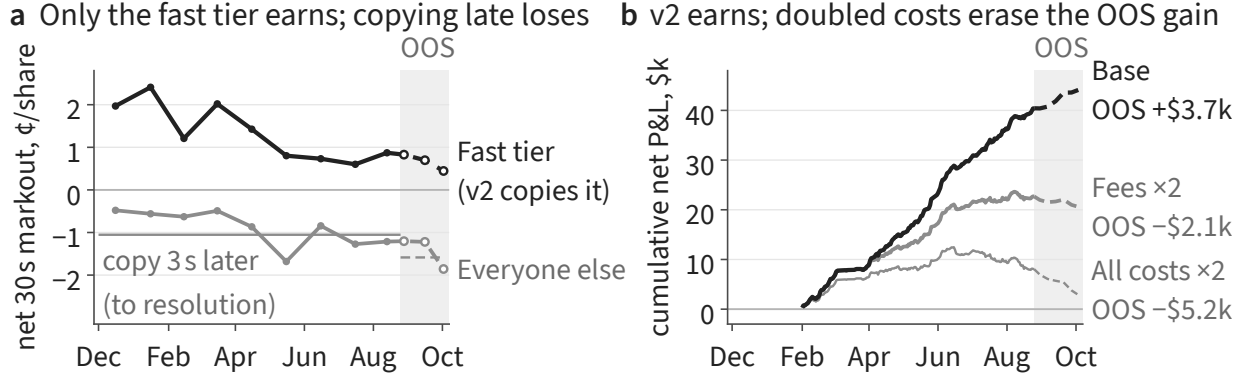


Figure 2. *Only the fast tier earns, and the edge is thin.* (a) Monthly net 30 s markout; thin grey: copying 3 s later, held to resolution. (b) v2’s equity curve, base and doubled costs. Source: results/v2.

next same-side print; fills past the tape’s end are rejected (1 IS, 1 OOS). New here: picking wallets by speed relative to the point, the sizing and limits, and the copier test.

Being fast ourselves. A CV model watches the video, calls the point early and sends a taker order before the stale quote is cancelled. We simulate it on past points with an assumed feed latency; all other parameters are measured. The order wins only if

$$V + \delta + D - \ell < L + R \iff V < L + R - D - \delta + \ell, \quad (1)$$

with V the feed delay, δ our pipeline plus network, D the 1 s hold, ℓ how early the CV calls, L the *stamp lag* (landing to official stamp) and R the reprice relative to the stamp (median -1.16 s), drawn from its measured spread once per tournament (the headline reading) or, as pre-registered, per point. Nobody publishes L : we *pre-registered* 2.0 s and also show 3.14 s, a *post hoc* estimate from one day. The *causal policy* trades every point plus the classifier’s false calls, each order chosen, signed and sized from what is known before it. We have no footage of the Polymarket matches, so the frozen classifier made no real calls on them: the policy draws calls from its measured table-tennis statistics (recall by lead, precision, 8.5 false calls an hour); fills use the post-point price and the live book, an upper bound. The *CV benchmark* trades only points the price later moved $\geq 4\text{¢}$ on, so it selects on the outcome.

Vision and speed. On broadcast tennis only out calls at the bounce met our 95% precision rule in training (test: 3 of 4 right 67 ms ahead). Our live engine makes about 30 phantom calls an hour outside play (8 with a rally gate): not yet safe to trade. With a simulated 1 s feed, our 54 ms pipeline, 65 ms of network and the hold, an order executes 2,119 ms after the point (p99 2,218 ms; not sent), within 3 s. A London server saw 99.74% of 287,577 book messages before a Florida one; with measured cold round trips (not order latency) the order executes in 2,089–2,146 ms (2,512 ms at worst).

Costs. Each match charges its own taker fee $f(q) = r q(1 - q)$ a share, 10,000 $r(1 - q)$ bps of notional (Polymarket, 2026a, fee schedule); today $r = 5\%$, 250 bps at 50¢. Fees paid were 120 and 179 bps of notional for v2 (IS, OOS) and 163–226 bps for the CV trader; fills cross the real spread (CV: walk the measured stale book). The stress doubles the fee (v2: also the spread).

How we ran the tests. The OOS period was evaluated once, on Oct 3 at 10:42 UTC, for the hypotheses and v1 book frozen in 9d61d1b; all later work knew that OOS, so its OOS figures are *non-blind*. Our experiment registry traces 2 strategy designs to OOS results (v2, after v1’s OOS loss; the per-wallet cap, after the OOS concentration; both fitted in sample), 4 presentation choices made after OOS cells existed (timing reading, post hoc lag, 1 s base case, 0.5 s main scenario; alternatives printed), 1 unexplained variant adoption and 4 mechanical corrections applied to every period (Appendix D).

5 Results

The copy book and overfitting checks. The fast tier earns every month and everyone else loses (Fig. 2a); at their own fills v2 earns in both periods, thinly, while the copier nets $-\$6,351$ IS and

Table 1. Net of costs, executable first: the copier and the causal camera policy (pre-registered lag) lose; v2 (the fast tier's own fills) is a benchmark, not attainable (CV on later moves: Table A2).

	\$ a day	¢ a share	Return	Volatility	Sharpe	Max DD	Turnover	¢, fees ×2
Executable: copier of v2 after the detection second, the venue's hold and the 1.98 s block lag								
Copier, IS	-\$31	-1.42	-194%	50.3%	-3.9	-114%	41	-2.07
Copier, OOS	-\$28	-1.02	-179%	55.9%	-3.2	-18%	11	-1.89
Causal selection: camera policy, every point, $L = 2.0$ s; simulated fills (upper bound)								
0.5 s, IS	-\$92	-0.33	-45%	5.2%	-9.0	-\$12.3k	68	-1.07
0.5 s, OOS	-\$146	-0.58	-114%	6.4%	-15.5	-\$6.5k	92	-1.53
1 s, IS	-\$124	-0.49	-65%	5.3%	-12.5	-\$14.4k	68	-1.24
1 s, OOS	-\$185	-0.87	-183%	8.0%	-20.7	-\$7.6k	99	-1.82
3 s, IS	-\$187	-0.99	-146%	6.9%	-20.7	-\$19.4k	73	-1.72
3 s, OOS	-\$230	-1.29	-280%	10.3%	-27.4	-\$9.0k	108	-2.24
Benchmark: the fast tier's own fills (v2; an event study, not attainable)								
IS	+\$196	+1.38	253%	17.5%	14.5	-2.0%	93	+0.77
OOS (non-blind)	+\$92	+0.60	148%	22.2%	6.7	-2.1%	129	-0.34

Notes: return, volatility (a year) and max DD on capital (3× peak dollars locked; causal CV max DD in dollars); turnover: dollars traded ÷ capital a year. ¢ CIs: copier [-2.31, -0.50], [-3.00, 0.84]; causal CV: 20-seed means, per-tournament timing, assumed feed latency; v2 [1.17, 1.59], [0.09, 1.13], Sharpe CI [11.9, 17.4], [1.9, 12.3], deflated Sharpe 0.997, 0.075, all costs ×2 +0.27, -0.84¢.

-\$1,104 [-\$3,221, +\$900] OOS. Six tests on daily P&L (probabilistic Sharpe and minimum track record against Sharpe 0 and 2, [Bailey and López de Prado, 2012](#); a Bonferroni haircut over all 4,219 variants, [Harvey and Liu, 2015](#); a 5-day block bootstrap): v2 passes all six in sample (haircut Sharpe 13.4) but 3 on the 40-day OOS ($t = 2.21$ vs 4.38). OOS, the CV benchmark fails all six at the pre-registered lag and passes 4 post hoc: none passes all six out of sample. Nearby settings work: all 55 sizing rules have an in-sample per-share CI above zero; the probability of backtest overfitting is 9–24%.

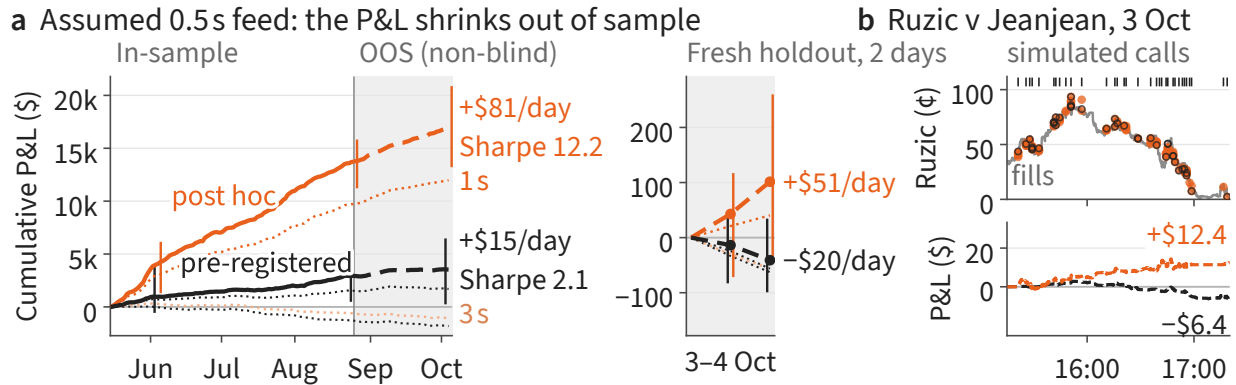
What speed is worth. Trading every point causally, the camera policy loses at every feed delay at the pre-registered lag (with R drawn per point: -\$136, -\$190 a day at 0.5 s). The post hoc 3.14 s lag earns at 0.5 s but loses OOS at 1 and 3 s (-\$45, -\$227); doubled fees make every causal cell lose. A replay of 9 live-recorded matches calling every point loses in 36 of 36 settings (-0.66¢ at 0.5 s). On points that later moved, the pre-registered benchmark breaks even at a 1.01 s delay OOS (Table A2); doubled fees erase it (best OOS -\$13): the gap is knowing which points matter.

If we were a quant firm with a licensed 0.5 s feed (Fig. 3; counterfactual: real prices, fills, fees and hold, simulated calls, no orders). Trading causally, it makes -\$92 / -\$146 a day (IS / OOS) at the pre-registered lag and +\$174 / +\$65 post hoc. On points selected by their later move: +\$15 / +\$81 OOS (pre-registered / post hoc), and -\$20.5 / +\$50.8 on a fresh holdout pre-registered before its data were fetched (16 matches, 2 days): anecdotal. Break-even feed price on those points: \$384 / \$2,397 a month (OOS), against quotes of \$1,250–\$10,000.

Pre-registered and blind tests. The one OOS evaluation with rules fixed in advance confirmed H6, an observational claim: the fast tier, picked on past months, kept a positive net 30 s markout out of sample (+0.67¢, 3/3 months) at its own fills; the trading rules failed (H1 -2.08¢; H2, H5 not significant; Table A1). Blind, frozen v2 made +1.22¢ [-0.19, 2.65] OOS on 11,307 unseen markets (before the timing fix): fail; run once on 108 never-fetched markets outside our universe (mostly doubles), the corrected copier made +8.80¢ [-34.41, 50.33] on 21 trades: fail, underpowered.

6 Risk management

Limits and the most we can lose. The net cap bounds the net position at \$95, not the loss: v2 buys both tokens in 79% of matches IS (71% OOS), so only the gross cap bounds a match's loss (\$1,479);



Counterfactual. Conditional benchmark. Real Polymarket prices, fills, fees and venue delay; calls simulated at an assumed feed delay (no feed purchased), only on the historical $\geq 4c$ jumps.

Figure 3. A firm with a licensed 0.5 s feed (counterfactual; trades selected on outcomes). (a) CV benchmark equity (20-seed mean, whiskers 10–90%; black pre-registered, orange post hoc, dotted 1 s, 3 s). (b) The showcase match fixed in advance: $-\$6.4 / +\12.4 (pre-registered / post hoc).

Table 2. Risk controls: operated in our backtests, or proposed for deployment (code: engine/risk).

Control	In the backtests	For deployment
Order $\leq \$1,000$; net ≤ 100 shares a match; price 0.05–0.95	operated (v2, CV)	engine code
\$3,000 gross a match	operated (v2)	proposed
CV early calls only at leads with precision ≥ 0.95	operated (CV)	proposed
\$1,000 daily loss stop	causal CV: never fired	engine code
Trailing edge $< 0.3¢$: half; ≤ 0 or 5% drawdown: stop	v2 IS replay: never fires	proposed
One confirmed false CV call halts the day	not run (wrong calls kept)	proposed
Stale feed (2 s) or vision (1 s), slow latency, kill	not applicable	engine code
Fee or hold change: no new risk until re-stressed	not run	proposed

the worst match made $-\$206$ IS, $-\$154$ OOS. The top five wallets carry 81% of v2's P&L IS, 142% OOS (129% with the per-wallet cap). Settlement, US access and tournament ticket terms add risk (Polymarket, 2026c,b; International Tennis Federation, 2026); no factor exposure (Appendix A8).

7 Liquidity and capacity

v2 holds its OOS edge up to $\$22,754$ of capital (at $5\times$ it loses), trading 0.07% of match volume; the CV benchmark's Sharpe halves at $\$40,000$ – $\$73,000$ post hoc (Fig. A6). With price impact fitted to 8.4 million IS orders, v2 at $1\times$ earns $\$88$ a day OOS, not $\$92$ ($\$71$ conservatively), and CV capacity moves under 1%; the simulation takes half the stale depth; other takers took only 4–11% of it in live books. After fixed data and running costs v2 makes $+\$29$ a day IS and $-\$75$ OOS: not yet a business.

8 Limitations and the out-of-sample rule

What earns is a benchmark or needs the post hoc lag; the camera policy uses table-tennis call statistics, not real calls on these matches, and upper-bound fills; L is unmeasured. Our one evaluation with rules fixed in advance used the pre-registered last 20% of matches: 2,617 over 38.7 days, 10.7% of the history. The organizers' rule takes the latest 20% by time (72.0 days from Jul 23), all read before the corrected policies were frozen; new matches alone would need 90 days. No policy here has an untouched test meeting the by-time rule; the frozen v1 test meets it only by match count, and every later OOS result is non-blind and exploratory. Next: a licensed feed to measure L and make real calls, then a 90-day forward test.

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Disclosure. We used no real money and connected no funded account. Simulated parts are labelled where they appear. We hold no licensed data feed, no footage of the matches we trade and no court access. Public datasets we did not build: the Match Charting Project (CC BY-NC-SA 4.0), the TrackNet broadcast tennis set and the OpenTTGames dataset. Open-source parts we did not build: the BlurBall and WASB model weights (Gossard et al., 2026; Tarashima et al., 2023), and the TrackNet weights and TennisProject/TennisCourtDetector code (yastrebkvs, 2026) (used unmodified, for evaluation only). The tennis rally is a Pexels stock clip (Gelato Prod, 2021); the logo on page 1 is the Gator Quant Hacks 2026 event’s own mark. Authors: Ojasva Mishra, Yoan Exposito, Rafael Penhas, Ian Hoang (University of Florida). Teammates’ work used here: Yoan Exposito ran the London latency probe on a Vultr server (sponsors/vultr), Ian Hoang re-derived 62 of 62 of our numbers in SQL on Snowflake and DuckDB, 20 recomputed from rows and 42 looked up (sponsors/snowflake), and Rafael Penhas built a Tiger Data tick store and live lab on the same tapes (sponsors/tigerdata).

Appendix A Calculations

The main pages stand alone. Below: formulas with worked examples (A), every test (B), extra figures (C) and the records (D). Each value is in results/paper/numbers.json with its source file.

A1. Sharpe ratio. $SR = \sqrt{365} \bar{d} / s_d$, where d_t is net P&L on every calendar day (zero on days without trades) and s_d its standard deviation. v2 IS: $\bar{d} = \$196.24$, $s_d = \$258.87$ over $T = 206$ days, so the daily ratio is 0.758 and $SR = 0.758 \times 19.105 = 14.48$. OOS: $\$92.19 / \$264.15 \times 19.105 = 6.67$.

A2. Sortino ratio. $\sqrt{365} \bar{d} / DD$, where the downside deviation is $DD = \sqrt{\frac{1}{T} \sum_t \min(d_t, 0)^2}$. For v2 OOS, $DD = \$125.64$, so the ratio is $\sqrt{365} \times \$92.19 / \$125.64 = 14.0$.

A3. Maximum drawdown. $MDD = \min_t (E_t - \max_{s \leq t} E_s)$, where E_t is cumulative P&L, as a share of capital. v2 OOS: $-\$469 / \$22,754 = -2.06\%$.

A4. Probabilistic Sharpe. The chance that the true daily Sharpe beats SR^* , given the sample length T , skew γ_3 and kurtosis γ_4 (Bailey and López de Prado, 2014):

$$PSR(SR^*) = \Phi\left((\hat{SR} - SR^*)\sqrt{T-1} / \sqrt{1 - \gamma_3 \hat{SR} + \frac{\gamma_4 - 1}{4} \hat{SR}^2}\right).$$

v2 OOS: $\hat{SR} = 0.349$, $T = 40$, $\gamma_3 = 0.326$, $\gamma_4 = 3.26$, so $PSR(0) = 0.987$.

A5. Deflated Sharpe. $DSR = PSR(SR_0)$, where SR_0 is the best Sharpe we would expect from N tries on pure noise, with γ the Euler–Mascheroni constant:

$$SR_0 = \sqrt{V} \left((1 - \gamma) \Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma \Phi^{-1}\left(1 - \frac{1}{Ne}\right) \right).$$

v2 OOS with $N = 3,386$ and $V = 1/(T-1)$: the two Φ^{-1} terms are 3.436 and 3.698, $SR_0 = 0.574$ a day (10.97 a year), and $DSR = 0.075$. Table 1 shows the lowest DSR over three choices of V . $N = 3,386$ is our variant count when the deflation ran; the count is now 4,219, which the haircut of Section 5 uses (at that N the OOS haircut is already 0).

A6. Probability of backtest overfitting. (Bailey et al., 2017). Split the daily P&L of 24 variants into $S = 16$ blocks. For each of the $\binom{16}{8} = 12,870$ ways to call half the blocks in sample, take the best in-sample variant, find its rank ω out of sample and set $\lambda = \ln \frac{\omega}{1-\omega}$; $PBO = \Pr(\lambda \leq 0)$. v2-safe grid: $PBO = 15\%$ (median $\lambda = 1.27$).

A7. Stationary block bootstrap. (Politis and Romano, 1994). Resample days in blocks of random length (mean 5 days) 10,000 times and read the percentiles. v2 OOS Sharpe: [1.9, 12.3], standard error 2.65, $P(SR \leq 0) = 0.0023$.

A8. Factor regression. $r_t = \alpha + \beta_M \text{MktRF}_t + \beta_S \text{SMB}_t + \beta_H \text{HML}_t + \beta_U \text{Mom}_t + \varepsilon_t$ on daily return on capital, with Newey–West errors (5 lags) (Fama and French, 1993; Carhart, 1997; Newey and West,

1987; French, 2026). v2 IS (138 days): $\alpha = 0.801\%$ a day ($t = 8.48$); $\beta_M = 0.173$ ($t = 1.36$), $\beta_S = 0.085$, $\beta_H = 0.044$, $\beta_U = 0.028$; $R^2 = 3.3\%$.

A9. Taker fee. $f(q) = r q(1 - q)$ a share at price q , or $r(1 - q)$ per dollar, which is 10,000 $r(1 - q)$ bps. At $r = 5\%$: $q = 0.5$ pays 1.25¢ (250 bps) and $q = 0.8$ pays 0.80¢ (100 bps).

A10. Point swing. $\Delta(k)$ of Section 2, with the match-win probability computed exactly by recursion over points, games, tiebreaks and sets (src/markov.py). Two equal players who win 0.645 of service points: at the first point $\Pr(\text{win}) = 50.0\%$, moving to 51.2% or 47.8%, so $\Delta = 3.4\%$. Serving at 5–5, 30–40 in the deciding set, $\Pr(\text{win}) = 33.7\%$, moving to 47.3% or 8.9%, so $\Delta = 38.4\%$. The mean $|\Delta|$ over a match is 5.7%.

A11. Latency budget. Eq. 1. With the pre-registered $L = 2.0$ s, $D = 1$ s and our measured $\delta = 0.119$ s, a point whose book reprices at the median $R = -1.16$ s needs $V < -0.28 + \ell$: no feed wins it without an early call. At the 90th percentile, $R = 0.08$ s and $V < 0.96$ s. The sweep's break-even (1.09 s IS) comes from these late reprices.

A12. Capital and return on capital. Capital is three times the peak dollars locked in open positions, each held 4 h or until resolution. v2 IS: peak \$9,434, capital \$28,302; return = $365 \bar{d}/\text{capital} = 365 \times \$196.24/\$28,302 = 253\%$ a year.

A13. Capacity and volume share. Capacity is the capital at which the seed-mean Sharpe falls to half its smallest-size value; share is our notional a day over market volume a day. CV, post hoc, IS, just below half Sharpe: \$10,428 a day against \$4.71M of in-play volume (0.22%) and \$117,047 traded by the fast tier 0–3 s after a point (9%).

Appendix B Everything we tested

Table A1. Every test we ran, its best result and our verdict. IS: in sample; OOS: out of sample (non-blind unless marked blind); ¢: net cents a share.

Test	What it was	Best result	Verdict
<i>Who is fast, and what they earn (public tapes)</i>			
H1 follow the jump	take the direction of a $\geq 4\text{¢}$ move, hold 30 s	OOS -2.08¢ $[-2.17, -1.98]$; 20 of 20 IS variants below zero	rejected
H2 favourites	buy in-play favourites priced 85–97¢	OOS $+0.80\text{¢}$ $[-1.36, 2.66]$	no edge
H4 score feeds	do public score feeds lag the book?	median lag 44.5 s ($n = 75$)	confirmed
H5 quote after jumps	post maker quotes after a move	OOS $+0.16\text{¢}$ $[-0.49, 0.87]$	not significant
H6 fast tier	wallets trading 0–3 s after a move, picked on past months	$+0.93 / +0.67\text{¢}$ net 30 s markout (IS / OOS prints only; $+0.75\text{¢}$ with August's IS prints); 9/9 and 3/3 months positive	confirmed
Kalshi lead-lag	Kalshi vs Polymarket on the same points	Kalshi first in 68.9% of 106,347 repricings (median 5 s)	lag mostly mechanical
Signal decay (TT5)	with-jump markout at 1–2 s minus 10–30 s after a move	OOS $+0.66$ $[0.45, 0.87]\text{¢}$; whole year $+0.19$ $[-0.26, 0.46]\text{¢}$; fast tier $+1.18$ to $+0.59\text{¢}$ (IS, 0 to 1–2 s)	decays in today's regime only
Clocks	when the book, the stamp and the tape move	book 1.2 s before the stamp ($n = 482$); tape 1.98 s late	measured
Factor regression	Fama–French three factors and momentum	$\alpha t = 8.48$; largest factor $ t = 1.36$	no factor exposure

Test	What it was	Best result	Verdict
<i>Copying the fast tier (v2 and relatives)</i>			
v1	the fast tier's own fills, dollar sizing (event study, not attainable); OOS evaluated once	IS +1.16¢; OOS +0.57¢, −\$36,056	failed
Sizing	55 sizing policies, walk-forward	G_50pct_net100: Sharpe 14.5 IS; all 55 with IS CI above zero (Sharpe 4.1–16.8)	adopted (v2); plateau
Exits	maker exit instead of holding (902 variants)	IS +0.53¢ [0.42, 0.64] (1 s / 5% regime); live fills adverse (27% filled in 30 s)	not adopted
Trade selection	which fast-tier trades to take (341 variants)	nested walk-forward +1.76¢ [1.36, 2.16] vs +1.16¢ (IS)	IS only
Side markets	side markets vs the match market (68 variants)	leaning maker +2.73¢ [1.73, 3.70] (IS); taking stale side quotes: no edge	IS only; tiny
v2	frozen copy book at the fast tier's own prices	Sharpe 14.5 / 6.7; OOS +0.60¢ [0.09, 1.13]	benchmark (Table 1)
v2, worse entry	enter ½ or 1 tick worse	IS +0.88 / +0.38¢; OOS +0.10 / −0.40¢	about zero OOS
v2, costs doubled	fees ×2 / all costs ×2	IS +0.77 / +0.27¢; OOS −0.34 / −0.84¢	fails OOS
v2, fixed costs	after central data and running costs	IS +\$29 a day; OOS −\$75 a day	fails OOS
v2, unseen markets	11,307 markets we never examined (blind)	IS +2.02¢; OOS +1.22¢ [−0.19, 2.65]	fail
v2-safe	net cap 50, fixed after v2's OOS losses	Sharpe 15.7 / 9.3; blind +1.21¢ [−0.02, 2.43]	fail (blind)
Per-wallet cap	\$1,000 a wallet a day plus retirement, chosen on IS from 12 variants, pre-registered	OOS +0.91¢ [0.20, 1.66], Sharpe 7.6, \$3,615 vs \$3,688; max DD −\$555 vs −\$469; top five 129% (was 142%)	pass (OOS, non-blind); a loss limit, backtest only
v2, strict timing	v2 on prints after the detection second, at the fast tier's own fills (observational)	IS +0.33¢ [−0.55, 1.21]; OOS +0.76¢ [−1.19, 2.65]	CIs span zero
Copier	those trades filled after the venue hold and the measured block lag, on the real tape	next same-side print: IS −1.42¢ [−2.31, −0.50], OOS −1.02¢ [−3.00, 0.84]; p90 lag: IS −1.44¢, OOS −1.05¢; last print before arrival (upper bound): IS −0.58¢, OOS +0.13¢ [−1.84, 1.96]	a copier loses
Copier, X1 (blind)	frozen copier on 108 never-fetched markets outside our universe (mostly doubles), pre-registered	+8.80¢ [−34.41, 50.33]; 21 trades, 11 matches, 9 days	fail; under-powered
Maker v1	pre-registered maker book (blind)	+1.87¢ [−0.14, 3.86] per fill, −\$379	fail

Test	What it was	Best result	Verdict
Overfitting checks PSR, MinTRL, haircut	deflated Sharpe, PBO, block bootstrap six tests incl. a Bonferroni haircut over 4,219 variants	DSR 0.997 / 0.075; PBO 15% (9–24% by block choice); $P(SR \leq 0) = 0.0023$ v2 IS 6 of 6 (haircut Sharpe 13.4); v2 OOS 3 of 6 ($t = 2.21$ vs 4.38); CV pre-registered 0, post hoc 4 OOS	luck not ruled out OOS no book passes all six OOS
Edge persistence	monthly decay, fee vs entrants, venue-rule breaks	net 0.16¢ a month down, before-fee −0.02¢ [−0.12, 0.07]; line hits zero in 3.3 months (0.6–12.8), a May step fits as well	cannot tell
Blind forward test	v2 on new matches, CV rule v3 as secondary	pre-registered but not run within the hackathon window (HYPOTHESIS_V2.md A5)	not run
<i>Being fast ourselves (computer vision)</i>			
H3 early call	table-tennis MISS call 50 ms before contact, precision ≥ 0.95	live causal engine: 4 of 41 misses, none wrong; offline evaluation (look-ahead feature): 11/11	precision passes; recall low
Spin model, table tennis	spin-aware early call, held-out footage	10/12 right at 50 ms vs the frozen model's 11/11 (both offline)	not adopted
Tennis tracker, real clip	public TrackNet detector with our linking and court-line fit on a Pexels rally (no labels)	ball in 89% of 300 frames; 48/48 spot-checks on the ball; court lines 0.36 px (median)	tracks; no in/out calls
Broadcast tennis	single-camera landing and out calls, TrackNet games 8–10 (tuned on 1–7)	out calls 5 of 5 at the bounce, 3 of 3 at 33 ms, 3 of 4 at 67 ms; landing 17 cm at the bounce, 0.7–1.2 m at 33–300 ms	bounce only at 95%; detector saw all games
Tennis spin model	spin-aware landing model (simulated physics)	0.57 vs 5.76 cm at 200 ms; OUT precision 1, recall 0.98	simulation only
Rally gate	a miss call trades only within 2.0 s of a bounce call (engine log; 0.5–1.2 s also run)	removes 5 of 7 calls on unlabelled balls; keeps 3 of 5 right ones (2 at 0.5–0.8 s)	not enough
GPU engine	streamed causal engine on one L4	119.9 fps; 4.6 ms p50; 0 frames dropped	measured
WebRTC pipeline	our clip over WebRTC into the engine	46 ms p50, 67 ms p99 frame to call; video leg 4–8 ms	measured
End to end	video frame to unsigned order on a live book	54 ms ours; 2,119 ms with a 1 s feed vs 3,000 ms	pass (order not sent)
Camera at the venue	CV benchmark (points that later moved) with a camera at the venue ($V = 0$)	+\$89 / +\$46 a day (pre-registered)	simulated; no camera
Latency sweep	CV benchmark, V from 0 to 60 s, six stamp-lag readings (944 cells)	pre-registered break-even 1.01–1.09 s	Table A2
CV benchmark, fees doubled	each fill pays its fee twice (Table A2)	every OOS cell loses (best −\$13 a day); fee 163–226 bps of notional	fails OOS

Test	What it was	Best result	Verdict
Fresh holdout, 0.5 s feed	CV benchmark (points that later moved) on 16 newer matches, pre-registered at dc95717	−\$20.5 (pre-registered) and +\$50.8 (post hoc) a day; CIs [−\$96, +\$57], [−\$38, +\$141]; at 1 s −\$28.1 / +\$20.2, at 3 s −\$31.4 / −\$28.3	anecdotal (2 days)
Price impact	log-size cost from 8.4 million IS taker orders; live-book walk	v2 1× OOS \$88 a day (conservative \$71, CI [−\$7, \$150]); others took 4–11% of stale depth	capacity holds; P&L thinner
CV rule v3	rule tuned on 360 IS variants	IS −0.22¢; OOS (non-blind) −0.52¢	fail
Causal camera policy	every point, calls from the classifier's measured statistics; simulated fills (upper bound)	0.5 s: −\$92 / −\$146 a day (IS / OOS); per-point timing −\$136 / −\$190; post hoc lag +\$174 / +\$65	loses at the pre-registered lag
Replay, all points	9 matches' real books, every point called	−0.94¢ [−1.28, −0.58] at V = 1 s; 36 of 36 settings below zero	loses
Replay, big points	only points with a large Markov swing (exploratory)	−0.95¢ [−1.64, −0.40]; 9 of 9 cells below zero	loses
Capacity	size grid for the CV benchmark and v2	post hoc Sharpe halves at \$40,000–\$73,000; pre-registered: none	small
Table-tennis markets	TT1–TT5 on Polymarket table tennis	no fast wallet qualifies; spread 94¢	untestable

Appendix C Robustness and extra figures

Table A2. *The CV benchmark on points that later moved* (Table 1's columns; trades only points the price later moved $\geq 4\text{¢}$ on, selected on the outcome; assumed feed latency (licensed feed not purchased); parameters measured; 20 seeds; †: ¢ CI spans zero). With R drawn per point, the post hoc reading loses (−\$17 a day OOS at 1 s).

	\$ a day	¢ a share	Return	Volatility	Sharpe	Max DD	Turnover	¢, fees ×2
Benchmark: CV on later moves , pre-registered $L = 2.0$ s; break-even V : 1.09 s IS, 1.01 s OOS								
0.5 s, IS	+\$28	+0.61	51%	11.9%	4.0	−4.5%	36	−0.20
0.5 s, OOS	+\$15	−0.02†	32%	12.3%	2.1	−3.8%	43	−1.04
1 s, IS	+\$15	+0.40†	30%	12.9%	2.1	−6.6%	31	−0.41
1 s, OOS	+\$4	−0.38†	11%	15.2%	0.3	−5.3%	38	−1.40
3 s, IS	−\$13	−1.35†	−57%	24.3%	−2.6	−24.8%	25	−2.14
3 s, OOS	−\$13	−1.69†	−98%	42.0%	−2.6	−19.0%	44	−2.71
Same, post hoc $L = 3.14$ s (CI 2.23–3.22 s); break-even V : 2.23 s IS, 2.14 s OOS								
0.5 s, IS	+\$133	+1.21	146%	9.4%	15.5	−1.3%	59	+0.41
0.5 s, OOS	+\$81	+0.82	123%	9.9%	12.2	−1.3%	68	−0.21
1 s, IS	+\$94	+1.11	118%	9.8%	11.9	−1.6%	51	+0.32
1 s, OOS	+\$57	+0.66	91%	10.1%	8.8	−1.6%	59	−0.37
3 s, IS	−\$6	−0.79†	−18%	16.8%	−1.4	−14.4%	23	−1.58
3 s, OOS	−\$11	−1.47†	−66%	31.7%	−2.4	−13.9%	37	−2.48

Table A3 lists the data sources; the figures support the main text’s pipeline, regime, replay, latency and capacity statements.

Table A3. *Real data behind every result; simulated: only the CV trader’s calls, their timing and its fills on the measured book, and the Hawk-Eye-class model.*

Source	What is real	Matches	Period	Used for
Polymarket books and trades (Polymarket, 2026a)	prices, fills, fees, delays	13,084 (8,554 ATP, 4,530 WTA); 1,310 at Slams (AO 415, RG 412, USO 483)	Oct 2025 – Oct 2026	fast tier; v2 (traded in 7,639 IS, 2,003 OOS); CV trader (7,755 matches at a 1 s hold; about 681 IS and 246 OOS traded a seed)
held back newest matches	same same	11,307 (10,855 ITF) 16 of 22	same Oct 3–4, 2026	blind test fresh holdout
Match Charting Project (Sackmann, 2026)	901,233 charted points	5,889 (3,337 men, 2,552 women)	2020–26	how points end (out 36–41%, net 25–26%)
TrackNet broadcast set (Huang et al., 2019)	19,835 labelled frames	10 (95 clips)		court, landing and out-call models
OpenTTGames (Voeikov et al., 2020); Pexels	120 fps video; phone video	41 held-out misses; one rally		early-call engine; tracker end to end

Frame to executable order in 2.12s; our part 54ms

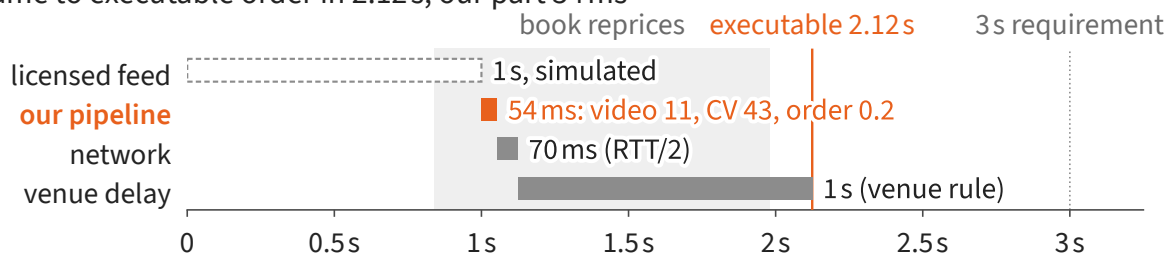


Figure A1. *We can trade inside 3s. Mean stage times over 24 runs on our own footage; the 1s feed is simulated; order built, not sent. Shaded: when the book reprices. Source: results/e2e.*

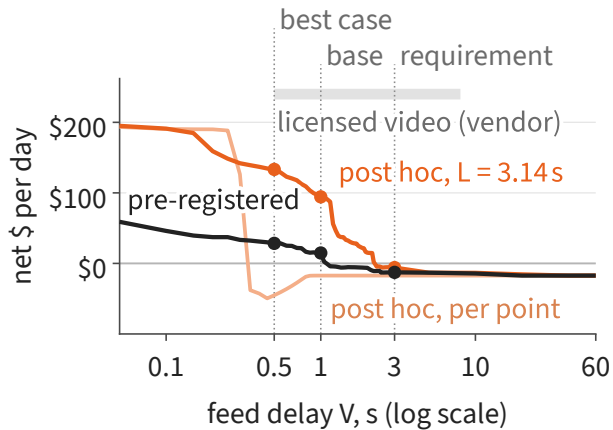
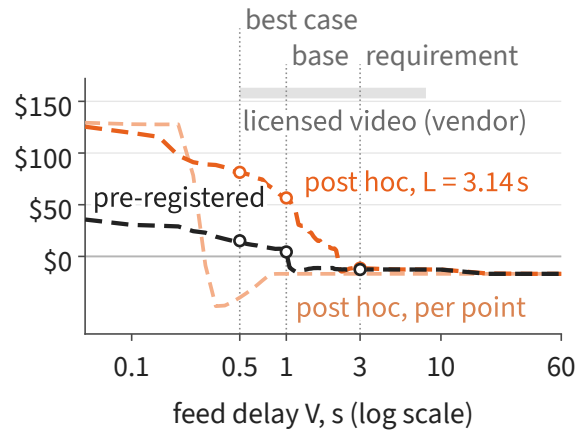
a IS: zero profit at 1.1–2.2 s of delay**b** OOS: zero profit at 1.0–2.1 s of delay

Figure A2. Each second of feed delay costs money. Simulated CV benchmark on points that later moved, (a) IS, (b) OOS; dots: Table A2; grey bar: vendor-stated licensed video. Assumed feed latency (licensed feed not purchased); parameters measured. Source: results/tier0.

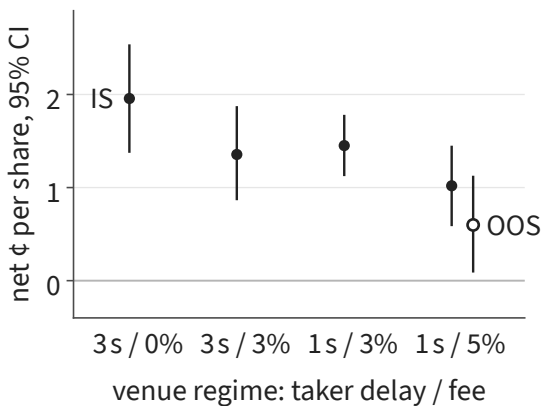
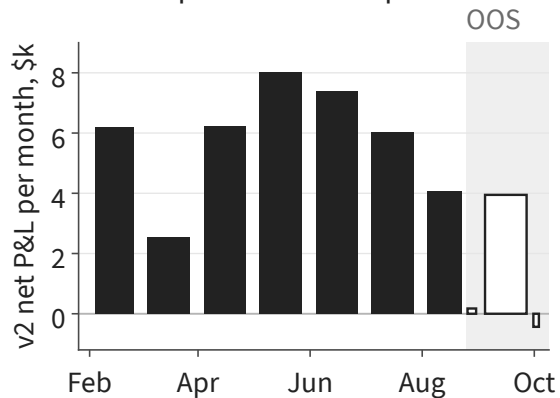
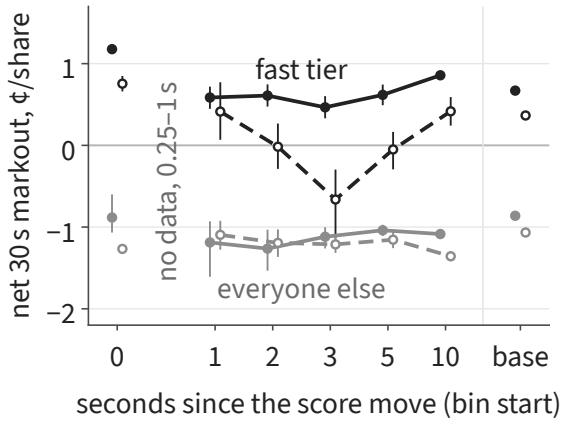
a Edge stays positive in every venue regime**b** 7 of 7 in-sample months are positive

Figure A3. v2's edge shrinks as the venue's protection shrinks, and stays positive in every regime. (a) Net \$ a share by venue regime (taker hold / fee), 95% CIs; (b) net P&L by month, at the fast tier's own prices. Source: results/risk/risk_stats.json.

a The fast tier stays ahead at every horizon



b Trading every point loses at every delay

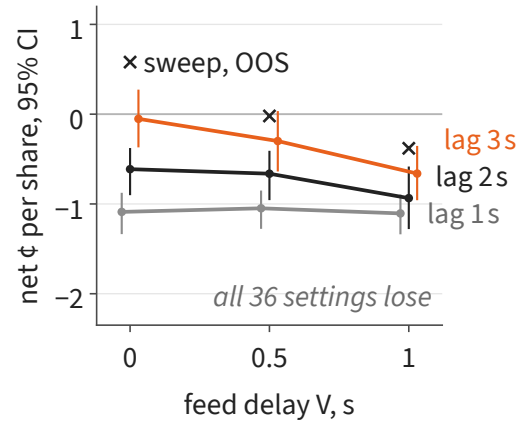
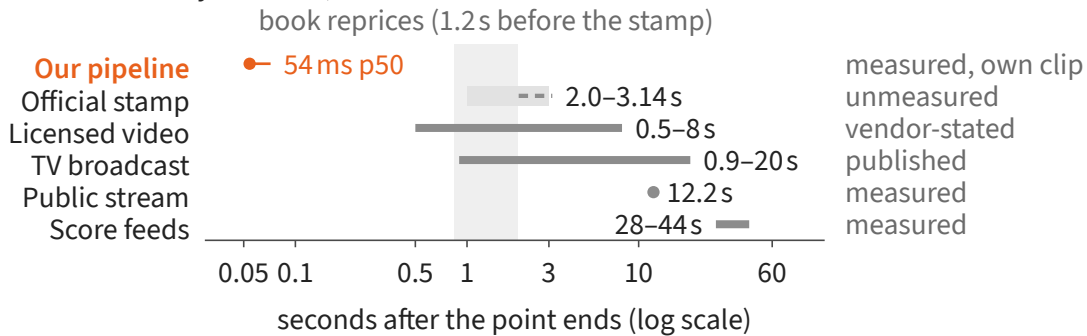


Figure A4. *The fast tier leads at every horizon; trading every point on real books loses.* (a) Net 30 s markout by seconds since the score move; filled: IS, hollow: OOS. (b) Replay of 9 matches recorded live on 2026-10-03 against their real order books, every point called; net ¢ a share marked at +30 s with match-clustered 95% CIs; ×: the sweep's OOS cell. One day, assumed feed latency, paper only. Source: results/{decay,replay}.

a Our order is ready in 54 ms; feeds take seconds



b Live engine: 4 of 41 misses called early, none wrong

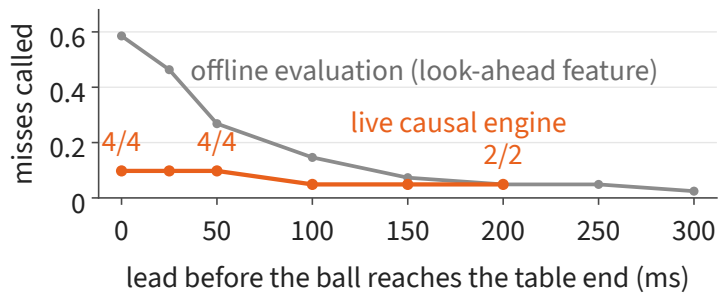
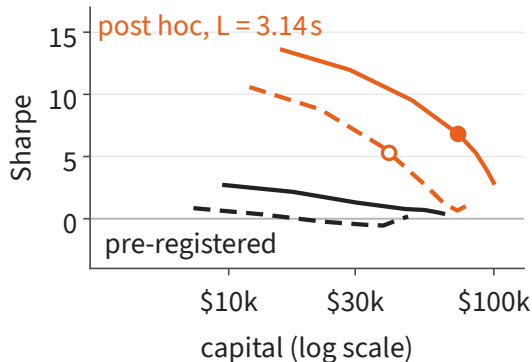


Figure A5. *Who sees the point first.* (a) Delay after the point ends (log scale): our pipeline (dot p50, whisker p99, own clip over WebRTC, laptop); the official stamp, dashed from the pre-registered 2.0s to the post hoc 3.14s over the cited 1–3s band (Regen Sports, 2026; Stats Perform, nd); vendor-stated licensed video (Stats Perform, nd; NEXT.io, nd; Sportradar, nd; Ably, nd); published stream delays (ISPreview, 2021; GL Systemhaus, 2018; The Desk, 2026); measured public stream and score feeds. Shaded: when the book reprises. (b) Early out-calls on 41 held-out table-tennis misses: the live causal engine (orange), which we can deploy, and the offline evaluation (grey), which used a look-ahead feature. Source: results/{e2e,webrtc,tracking,engine}, research/v2/latency.

a Sharpe halves by \$40k–\$73k (post hoc)



b The cheapest data stack needs \$14k–\$16k

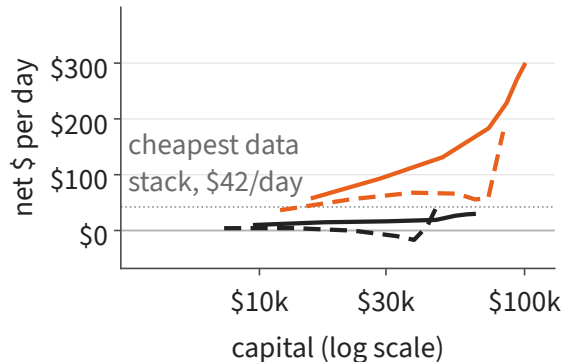
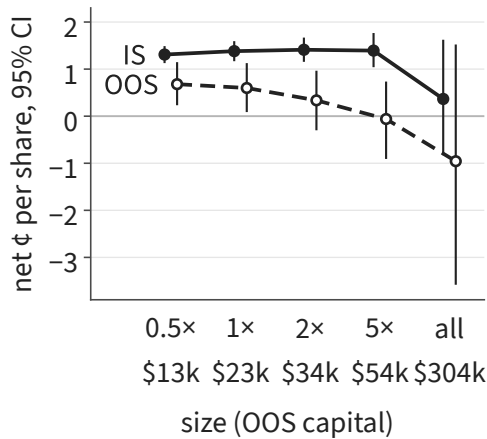


Figure A6. *How much capital the CV strategy can use at a 1 s feed.* (a) Sharpe and (b) net \$ a day against capital (net cap 50 to 5,000 shares, \$250 orders, 10 matches a day, half the stale depth); circles: where the post hoc Sharpe halves; dotted: the cheapest data stack. Solid: IS; dashed: OOS. Simulated, assumed feed latency (licensed feed not purchased); parameters measured. Source: results/capacity.

a v2's OOS edge fades as size grows



b After fixed costs only v2 IS stays positive

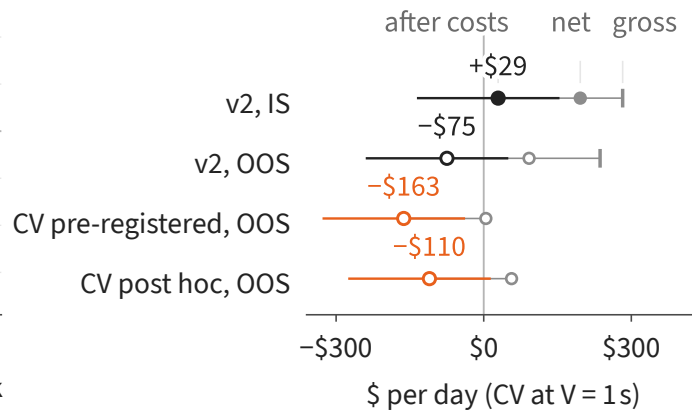


Figure A7. *v2's capacity is small and fixed costs bind.* (a) Net edge by size (caps scaled; optimistic at large size, since we model no price impact); filled: IS, hollow: OOS. (b) \$ a day gross, after fees and after central fixed costs (bar: low to high); CV rows at the 1s base case, OOS, pre-registered first; assumed feed latency (licensed feed not purchased); parameters measured. Source: results/financials.

Appendix D Records

The full logs are in the repository.

Pre-registration. H1–H4: HYPOTHESIS.md, committed at 7232986 (2026-10-03 09:50 UTC) before any result. H6, v2 and the forward test: HYPOTHESIS_V2.md (amendments A1–A5). Table tennis: HYPOTHESIS_TT.md. The CV simulation's research/v2/tier0/PREREG.md says it was written at 17:25 UTC; it was committed with its results at 19:24 UTC (a5769c7), and the held-out read log shows the OOS grid first ran at 17:29 UTC. git log on these files lists 19 commits.

Choices made after an OOS look. DEVIATIONS.md and research/v2/*/DEVIATIONS*.md give every rule change and its reason. The experiment registry (results/provenance/experiments.json, 68 events with times and commits) classifies them (Section 4 gives the classes). Every decision made after an OOS look, in its words, each printed once. Defect fixes, applied to every period: (1) After the single test read, a label audit relabelled 20 of 41 test misses; the frozen model's offline evaluation then scores 8/8 instead of 11/11; (2) To remove onset hindsight, v2 adopted causal window T3e after seven variants had OOS values; only T3e's interval excluded zero; no selection reason was recorded; (3) Capital lock-up was set to an ex-ante 4 hours after an OOS look; per-share P&L and Sharpe did not change; (4) A look-ahead in CV camera coverage was fixed after the first CV P&L, including OOS, existed; the fix lowered coverage; (5) Five CV model fixes from the verifiers were adopted after OOS P&L existed; two lowered P&L and three changed it little; (6) Capacity-study defects found by a verifier were fixed after the OOS capacity grid had run; capacity estimates moved down slightly; (7) After review, v2 and its copier were re-timed to use only prints after the detecting print's second; the copier's estimates stay negative; (8) After review, simulated CV trades were taken on every point from the call model's calls, not only on later price jumps; this lowered simulated P&L. Other decisions: (9) v2 was designed after v1's OOS loss was known; its parameters were fitted on in-sample data only; (10) Forward-test primary metrics were redefined after a dry run on OOS days; the forward test was never run; (11) The v2-safe variant was designed with v2's in-sample and OOS loss profiles known; it was selected in sample by a pre-declared rule; (12) The per-tournament

CV timing reading was made the headline after OOS results existed; it raised OOS P&L over the pre-registered per-point reading; (13) The CV headline was switched from the pre-registered model to the corrected model after OOS results existed; this lowered it about 3-5x; (14) The calibrated 3.14 s stamp-lag reading was made the post hoc CV headline after OOS sweep cells existed; it raised OOS P&L; (15) The 1 s feed-delay base case was fixed after the latency sweep's OOS cells had been computed; (16) The per-wallet cap was designed because the OOS showed five wallets carried 142% of v2's OOS P&L; it was selected in sample; (17) The licensed 0.5 s feed was made the main latency scenario after the latency sweep's OOS cells had been computed. Result labels: Executable copier: OOS, non-blind; copier entering after the fast tier at our latency; Live-book replay: Exploratory replay on recorded books (the same matches built the model's inputs); Causal CV selection: OOS, non-blind; CV trades chosen before each point from the call model; simulated fills are an upper bound; Fast tier's own fills (v2): In sample (rule designed after the v1 OOS result; causal variant adopted after an OOS look); Fast tier's own fills (v2): OOS, non-blind (period read before this design; variant chosen after an OOS look); CV on later moves, pre-registered lag: OOS, non-blind; conditional on realised jumps of at least 4c; stamp lag 2.0 s pre-registered, timing reading chosen after an OOS look; CV on later moves, post hoc lag: OOS, non-blind; conditional on realised jumps of at least 4c; post hoc stamp lag 3.14 s.

Held-out data. results/oos_peek.log records each run that read held-out data, including reproductions, verifier runs and replays; the registry counts 10 pre-registered evaluations and 12 reproductions.

Variants. 4,219 in all (results/paper/variants.json): H1-H6 pre-registered and H5/H6 rules (44); v2 lenses (exit, sizing, selection, cross-market, latency, Kalshi) (3,342); v2-safe grid (24); tier-0 pre-registered scenarios (432); tier-0 v3 in-sample grid (360); maker v1 evaluated variants (8); cost-stress cases (4); table-tennis tests TT1-TT5 (5). The fresh holdout, the fees-doubled CV cells and the PSR tests choose nothing and add no variant. Sensitivities that chose nothing: 944 latency-sweep cells and 36 replay cells, plus 4 rally-gate values (Section 4). The per-wallet cap of Section 6 added 12 more variants after this count.

Grids and sources. Latency sweep: results/tier0/latency_sweep.json; replay: results/replay/selective/selective.json; fixed costs: results/financials/financials.json; latency sources: research/v2/latency/results.json; every number with its source: results/paper/numbers.json.

Forward test and live session. Blind forward test: pre-registered but not run within the hackathon window (HYPOTHESIS_V2.md A5). The live paper session was stopped by a team decision and is not used (research/v2/maker/DEVIATIONS_LIVE.md, L15).